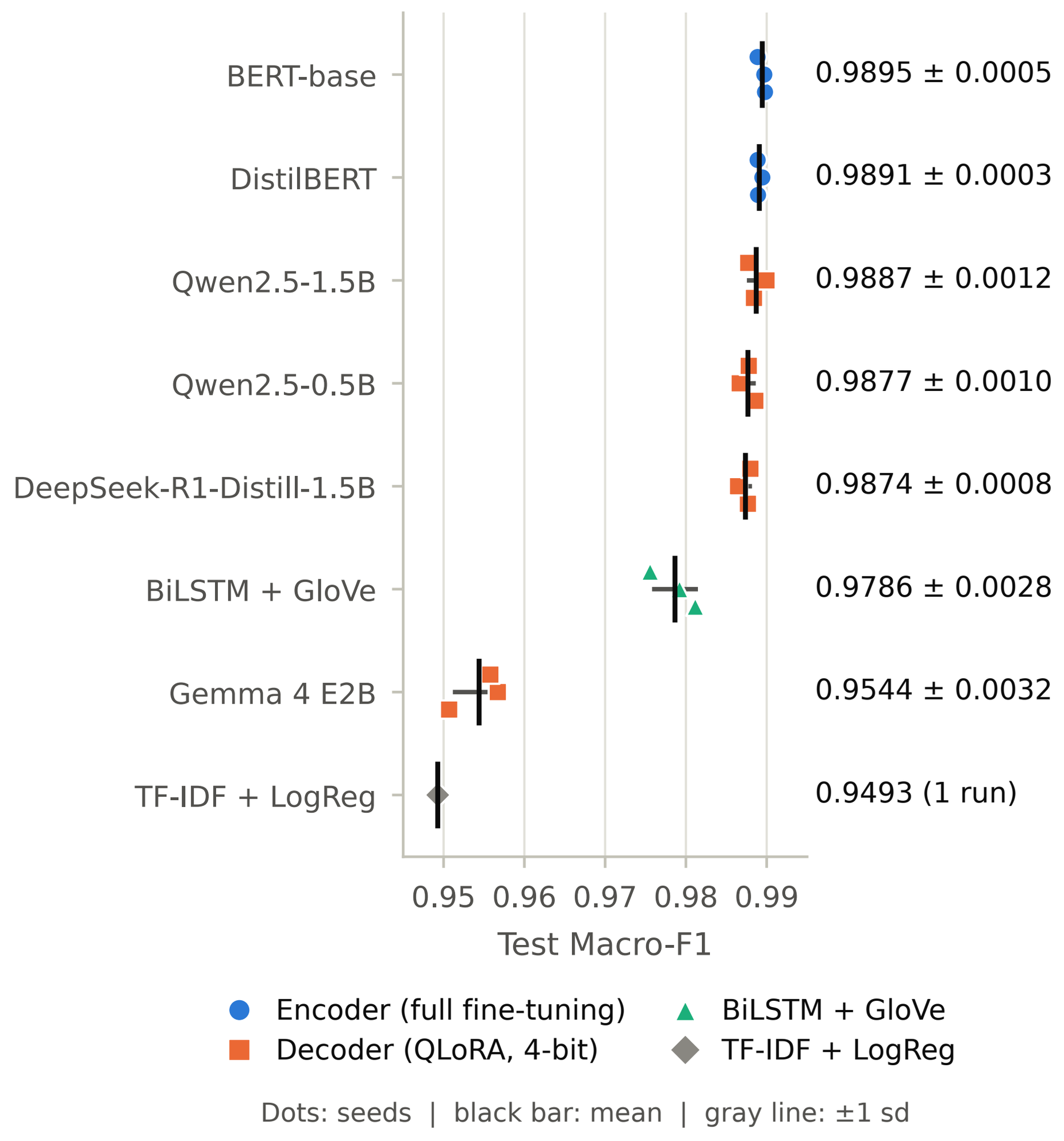
BERT-base
DistilBERT
Qwen2.5-1.5B
Qwen2.5-0.5B
DeepSeek-R1-Distill-1.5B
BiLSTM + GloVe
Gemma 4 E2B
TF-IDF + LogReg
0.9895 ± 0.0005
0.9891 ± 0.0003
0.9887 ± 0.0012
0.9877 ± 0.0010
0.9874 ± 0.0008
0.9786 ± 0.0028
0.9544 ± 0.0032
0.9493 (1 run)
0.95
0.96
0.97
0.98
0.99
Test Macro-F1
Encoder (full fine-tuning)
Decoder (QLoRA, 4-bit)
BiLSTM + GloVe
TF-IDF + LogReg
Dots: seeds | black bar: mean | gray line: ±1 sd